MIT SCHOOL OF COMPUTING
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A Leap Towards World Class Education

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Subject: IDS Roll No.: 27 Exp. No.: 02
Name of the Experiment:

Marks	Teacher's Signature with date
10	

Performed on:
Submitted on:

Q.1. Define sequential organization list the characteristics of linear data structure that used sequential organization.

→ Sequential organization means storing data elements one after another in continuous memory location. Each element is directly accessible using its index.

Characteristics :—

1 Elements are arranged in sequence
2 Stored in contiguous memory location
3 Direct access using index is possible
4 Traversal is easy
5 Insertion and deletion require shifting elements

eg: Arrays and Lists

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Q.2. Explain the concept of array as an Abstract Data Type (ADT). Components are Basic operations.

→ An array ADT is a logical model that defines stored in contiguous memory location.

(a) Components :- 1) Data element
2) Index
3) Size.

(b) Basic operations :- 1) Traversing
2) Insertion
3) Deletion
4) searching
5) Updating

Array ADT hides implementation details and focuses on operations that can be performed on the data.

Q.3. Demonstrate how insertion, deletion and searching operation are performed on a one-dimensional array with suitable example:

→ Insertion : To insert an element, space must be available. Elements after the position are shifted to the right. Time complexity is O(n)

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Deletion : To delete an element it is removal and remaining elements are shifted left. Time complexity is O(n)

Searching : (1) Linear search checks each element sequentially [O(n)]

(2) Binary search works on sorted arrays and is faster [O(log n)]

Q.4. Analyse the advantages and limitations of using arrays for dates storage. Compare array operations such as insertion, deletion and searching based on their efficiency.

→ Advantages : —

- Fast access using index [O(1)]
- Simple to implement
- Efficient memory usage.

Limitations : —

- Fixed size
- Insertion and deletion are costly
- May cause memory wastage.

Efficiency comparison —

- Access → O(1)
- Insertion → O(n)
- Deletion → O(n)

Searching :—
O(n) or O(log n)

Q.5. Design a two dimensional array structure to store and process student academic data. Explain how much the data can be accessed and manipulated efficiency. Extend your discussion to the use of n-dimensional array in real world applications.

→ A two dimensional array stores data in rows and columns. Rows can represent students and columns can represent subjects.

Examples = Marks [i] [j]
where,
i = represents students numbers and
j = represents subject.

This allows easy access using rows and columns index.
Extension to n-dimensional arrays
- Used in 3D graphics
- Used in image processing
- Used in scientific computations
- Used in machine learning and data analysis

Multi-dimensional array help in representing complex real-world data efficiency.